Lesson notes · for you

A language model, and changing one

AI · University · Lesson 2

Materials
A phone with Darsi; Paper for notes; The slides from the kit

Learning goals

- A guess, not a lookup
- Unless a tool searches

Activity: In the lab

1. Read the 16 example conversations for the new bus tool, choose how many old conversations to mix in, and press “Teach”. You can stop at any time.
2. Say what stayed frozen (the embedding, the output layer and the lower blocks) and what learned (the top block).
3. Press “Test it” and read both lines: the new tool with villages it never saw, and the old skills. If there is time, teach again with no old conversations mixed in, and compare.

Discussion

- What did not move?

Check yourself

- Fine-tuning here changed which block?

Try it this week
Write what the small fine-tune changed here, and what it was too small to change.

Worksheet

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Name _____ Class _____ Date _____

1 **Does a language model look facts up as it writes?**

- ◯ No: it guesses the next token from patterns, unless a separate tool searches
- ◯ Yes: it looks each fact up in the text it was trained on
- ◯ Yes: it searches the internet for each fact as it writes

2 **In the teach lab, which parts of Darsi Tiny stay frozen?**

☐ The embedding ☐ The lower blocks ☐ The top block ☐ The output layer

3 **Match the choice to what it does.**

Freeze everything but the top block	• • • •	An old skill can slip
Mix in no old conversations	• • • •	The old skills are kept
Mix in many old conversations	• • • •	The change stays small and cheap

4 **Order a small fine-tune.**

- ☐ Teach the top block with a few examples
- ☐ Start from a model that already exists
- ☐ Test the new job and one old skill
- ☐ Freeze everything but the top block

✎ **What stayed frozen?**

Check your answers

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1 **No: it guesses the next token from patterns, unless a separate tool searches**

It writes one token at a time from patterns stored in its weights.

2 **The embedding; The output layer; The lower blocks**

Only the top block learns, with the small normalising step after it.

3 **Freeze everything but the top block — The change stays small and cheap**
Mix in no old conversations — An old skill can slip
Mix in many old conversations — The old skills are kept

Say which weights moved, and retest an old skill, before you claim the model learned a job.

4 **Top to bottom: 3, 1, 4, 2**

Start from a trained model, freeze most of it, then teach the rest, and test before you trust it.